Anastasios A. Manganaris

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EDUCATION

Purdue University, West Lafayette, Indiana — May 2028
Ph.D. Computer Science — GPA: 3.78 (4.0 scale)

Courses: Algorithms, Introduction to Robot Learning, Operating Systems, Artificial Intelligence, Imitation Learning, Optimization

The Ohio State University, Columbus, Ohio — May 2023
B.S. Computer Science & Engineering, specialization: A. I. — GPA: 3.904 (4.0 scale)

Courses: Calculus, Linear Algebra, Differential Equations, Algorithms and Data Structures, Computer Systems, Web Application Development, Techniques in A.I., Computer Architecture, Neural Networks, Machine Learning Theory, Programming Languages, Database Systems, Parallel Computing, Computer Vision, Automata & Formal Languages, Compiler Design and Implementation, Realtime Rendering

Anderson High School, Cincinnati, Ohio — June 2019
Graduated with Honors — GPA: 4.663 (4.0 scale)

EXPERIENCE

Research Assistant at Purdue University — May 2024 - Present
Researching formal methods for more general and safe robot policy learning, multi-agent task and motion planning, and hierarchical reinforcement learning.

Teaching Assistant at Purdue University — August 2023 - May 2024
Instruction, course organization, development, and grading for Data Structures and Algorithms (CS251, Head TA for CS253)

Research Assistant at The Ohio State University — December 2021 - May 2023
Investigated identifying RNA chemical modifications from Nanopore sequencing data and relationship between Nanopore sequence quality and RNA structures.

Software Developer at SentriLock, LLC — June 2018 - August 2018
Designed, implemented, and deployed an application for an analytics dashboard.

PUBLICATIONS

A. Manganaris, J. Lu, A.H. Qureshi, S. Jagannathan. Graph-of-Constraints Model Predictive Control for Reactive Multi-agent Task and Motion Planning, IEEE RAS International Conference on Robotics and Automation (ICRA), 2026.

A. Manganaris, V. Giammarino, A.H. Qureshi, S. Jagannathan. Formal Methods in Robot Policy Learning and Verification: A Survey on Current Techniques and Future Directions, Transactions on Machine Learning Research (TMLR), Dec 2025.

A. Manganaris, V. Giammarino, A.H. Qureshi. Automaton Constrained Q-Learning, Neural Information Processing Systems (NeurIPS), 2025.

SKILLS

Programming Languages: Python, C/C++, MATLAB, Java, Rust, Common Lisp, GNU Guile, Haskell, Ruby, JavaScript, PHP

Software & Services: ROS2, ROS, JAX, PyTorch, Linux, Docker, AWS, Slurm, OpenMP, MPI, Emacs, Git, Rails, ReactJS, Nix, and GNU Guix.